SAR Lab 7 Report: Xinmo Landslide, Sichuan Province, China

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1 Objective

This report summarizes the SAR processing completed for Lab 7. The case study is the Xinmo/Maoxian landslide in Sichuan Province, China, which failed on 2017-06-24. The lab asks for a Sentinel-1 SAR time-series analysis using COMET-LiCS interferograms, followed by an assessment of whether coherence loss and Sentinel-1 VV backscatter change are useful for mapping the landslide.

The workflow completed here used pre-processed COMET-LiCS interferograms, subset them to the landslide area, converted them to MintPy input files, and ran SBAS time-series processing. The available local LiCSAR stack is pre-event only: the latest interferograms end on 2017-06-07, which is 17 days before the landslide. This is important for interpretation: the MintPy results are useful for examining possible pre-failure motion, but the local coherence raster cannot directly show event-spanning coherence loss for the 2017-06-24 failure.

2 Data And Processing

The SAR input data were downloaded from the COMET-LiCS frame `062D_05831_131313`. Interferograms were selected before the landslide date with a maximum temporal baseline of 72 days, then converted into a MintPy stack. MintPy was run using SBAS network inversion to estimate displacement time series, velocity, temporal coherence, and related quality-control products.

The Google Earth Engine notebook was also used to compare Sentinel-1 GRD VV backscatter before and after the event. The pre-event composite used 9 Sentinel-1 acquisitions between 2017-05-02 and 2017-06-19. The post-event composite used 10 acquisitions after the event, including dates from 2017-06-30 through 2017-08-12. This means the VV-change product is a true pre/post-event comparison, unlike the currently available local LiCSAR coherence file.

3 MintPy SBAS Results

3.1 Line-of-sight velocity

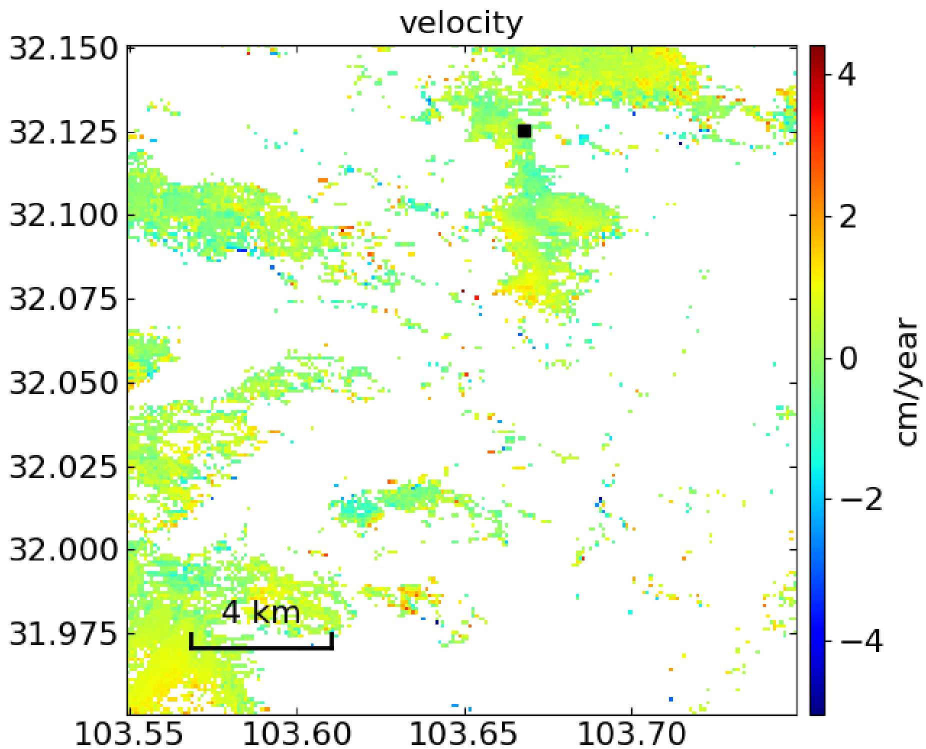
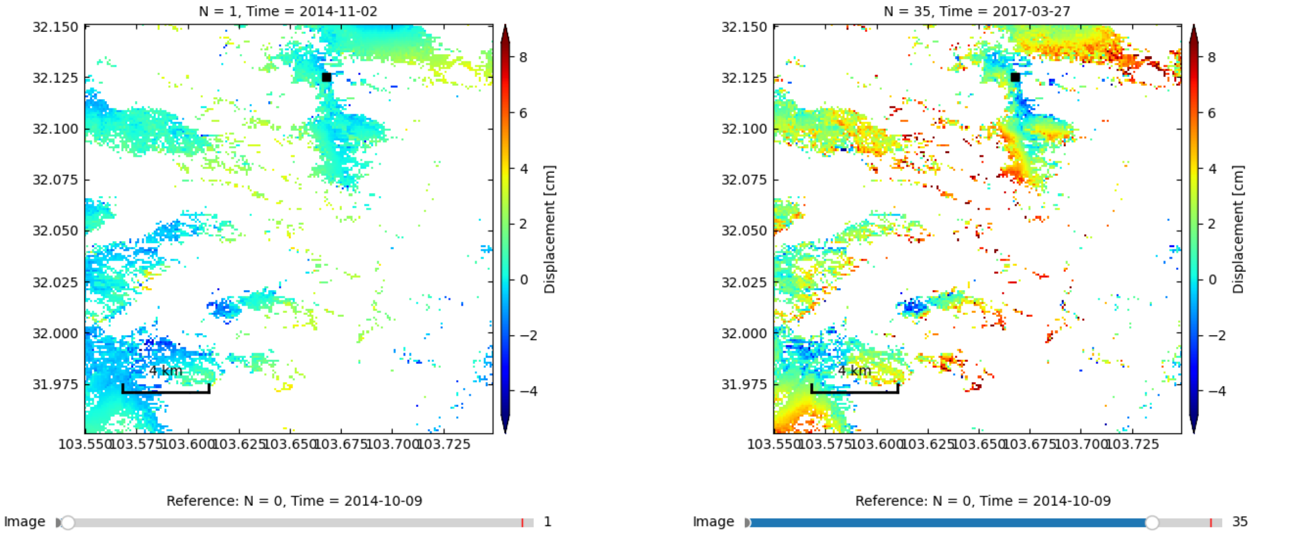


Figure 1: MintPy velocity map from the processed pre-event LiCSAR stack. The color scale is in cm/year.

The velocity map shows mostly small line-of-sight rates across the subset, with a localized positive displacement pattern near the central to northern part of the scene. The strongest coherent signal is on the order of a few cm/year. This is consistent with the possibility of slow pre-failure slope motion before the 2017-06-24 landslide, although the result is not spatially complete because many areas are masked or decorrelated.

The black square marks the selected reference area. The choice of reference point matters in this lab because the landslide region and nearby terrain can contain seasonal or unstable signals. A reference point close to, but outside, the unstable slope helps reduce broad atmospheric or seasonal artifacts while preserving the local deformation signal.

3.2 Displacement snapshots



(a) Early displacement snapshot, 2014-11-02.

(b) Later pre-event displacement snapshot, 2017-03-27.

Figure 2: Selected MintPy displacement snapshots relative to the 2014-10-09 reference epoch.

The early snapshot is close to the reference epoch and shows relatively small displacement patterns. By 2017-03-27, several areas show cumulative displacement of several centimeters. The central/northern slope region near the marked reference area has a stronger positive signal by this time. This supports the interpretation that the slope system experienced measurable pre-event deformation before the landslide, but the spatial pattern is patchy and should not be interpreted as a precise landslide boundary.

3.3 Full MintPy time series

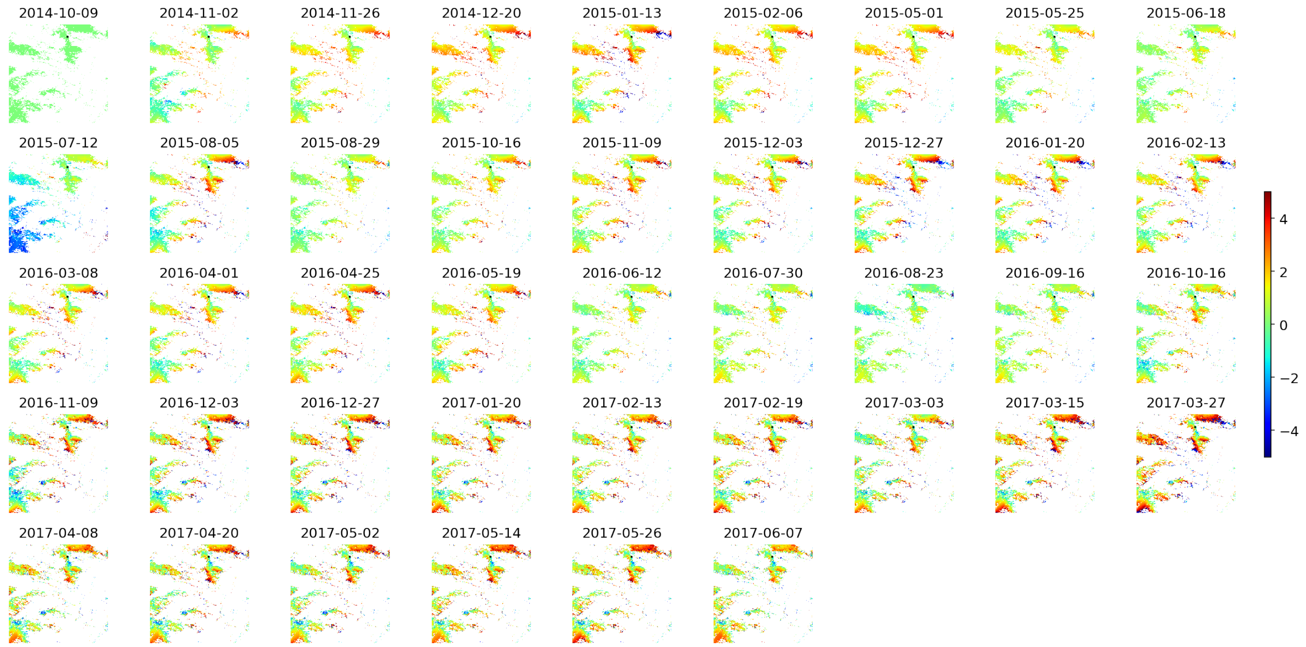


Figure 3: MintPy displacement time series from 2014-10-09 to 2017-06-07. The stack ends before the 2017-06-24 landslide.

The full time-series panel shows that the displacement field evolves over multiple years before the event. The latest available date is 2017-06-07, so the final failure and immediate post-failure deformation are not captured in this MintPy stack. The time series is therefore best interpreted as evidence for pre-failure motion and not as a direct map of the landslide scar or deposit.

4 Quality Control And Reliability

4.1 Temporal coherence

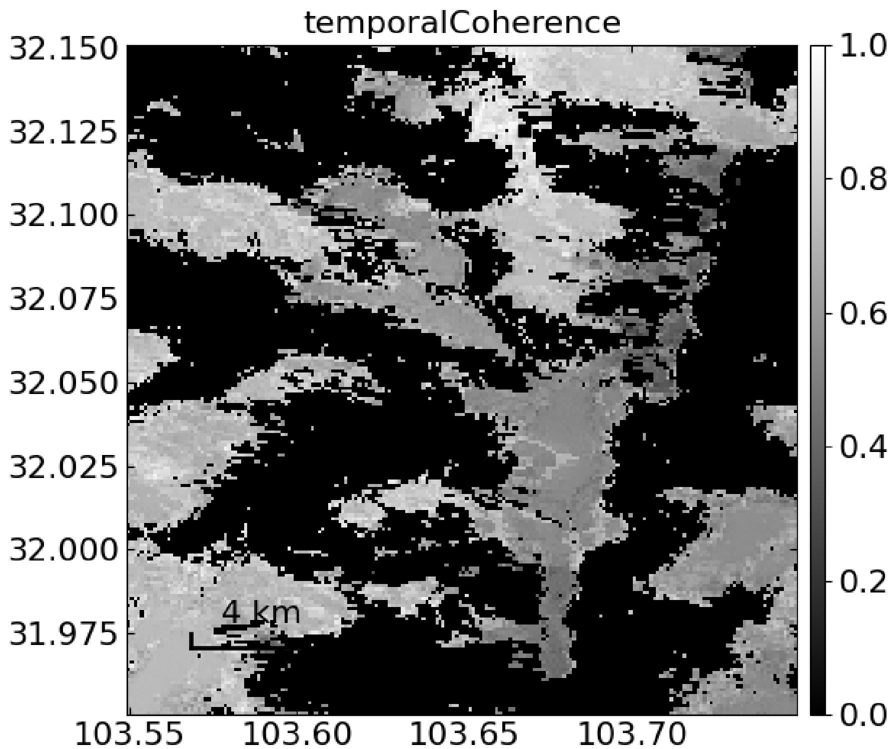


Figure 4: Temporal coherence from MintPy. White pixels are more reliable; black pixels are low-coherence or masked.

Temporal coherence is highly variable across the mountainous scene. Reliable pixels occur in patches, while large areas are low-coherence or masked. This limits the confidence of the velocity and displacement maps in vegetated slopes, radar shadow, layover, snow-affected areas, and other decorrelated terrain. The velocity result should therefore be discussed only where temporal coherence is moderate to high.

4.2 Average spatial coherence

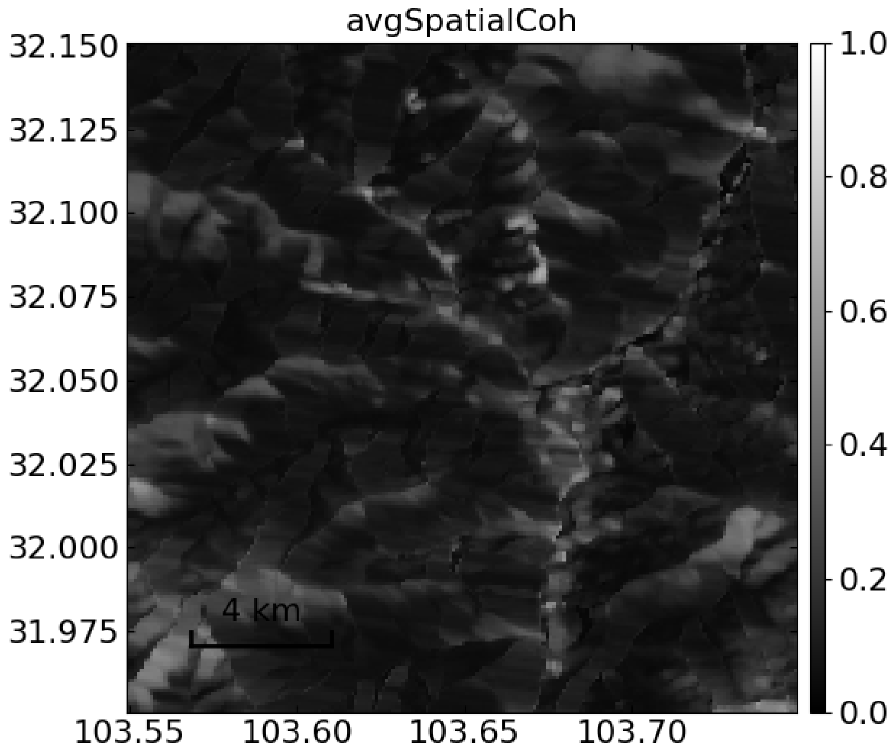


Figure 5: Average spatial coherence over the interferogram stack.

The average spatial coherence is generally low across much of the subset. This is expected in steep terrain with vegetation and complicated radar geometry. It explains why coherence-based classification is difficult here: low coherence is not unique to the landslide, and many non-landslide areas can also appear dark because of vegetation, shadow, layover, or temporal decorrelation.

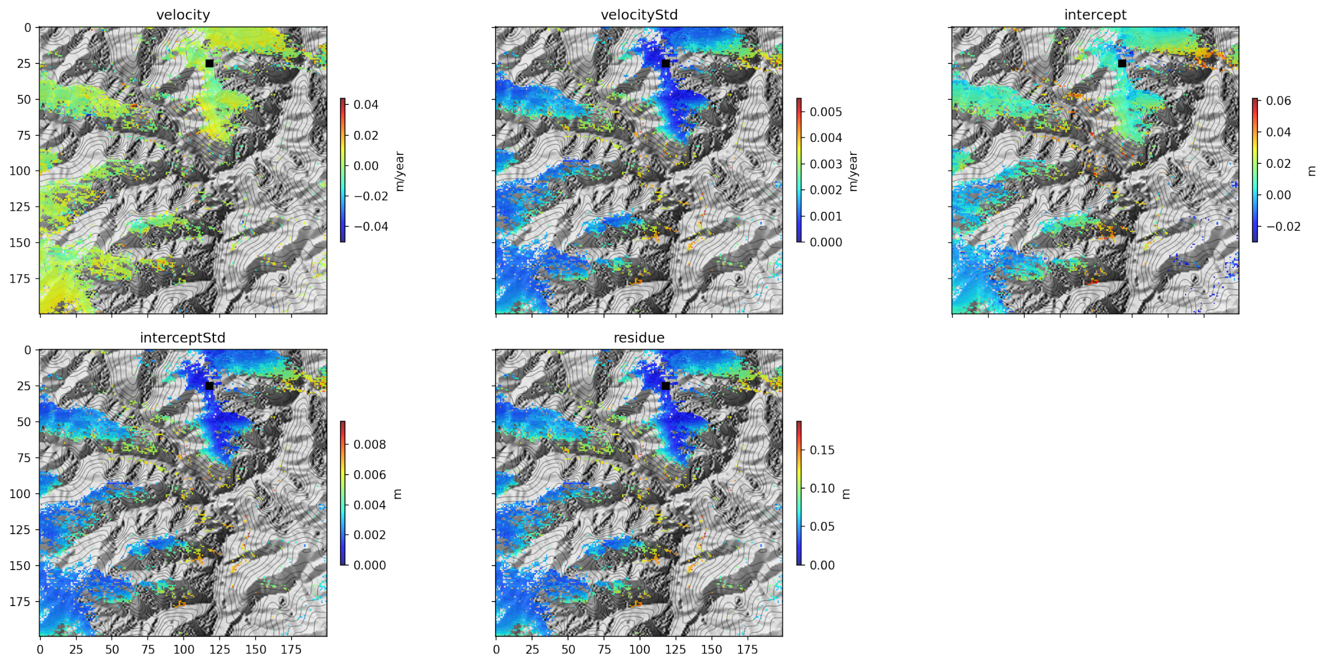


Figure 6: MintPy velocity product with standard deviation, intercept, and residual panels. These diagnostics show where the velocity estimate is more stable and where residuals or uncertainty are higher.

The MintPy diagnostic panels reinforce the same conclusion. Some areas have relatively low velocity uncertainty, but other parts of the scene contain higher residuals and limited valid pixels. The displacement signal near the slope of interest is meaningful as a pre-event indicator, but it should be treated cautiously because the mountainous setting produces spatially uneven InSAR quality.

5 Coherence-Loss Mapping

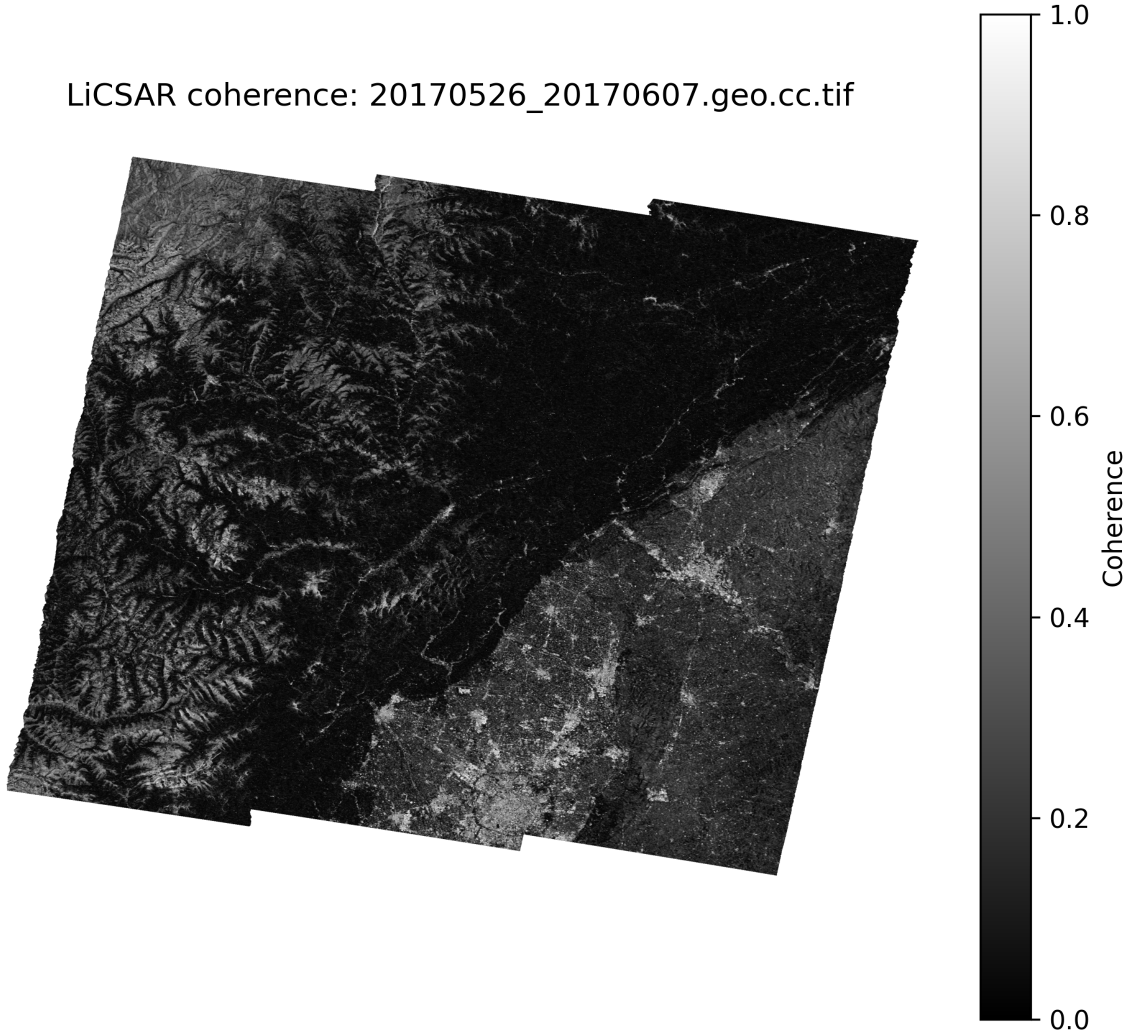


Figure 7: Closest available local LiCSAR coherence raster: `20170526_20170607.geo.cc.tif`. This file does not span the 2017-06-24 event.

The lab asks whether coherence loss is useful for mapping the landslide. In principle, an interferogram that spans the landslide date could show strong coherence loss where the ground surface changed abruptly. However, the available local file shown above covers 2017-05-26 to 2017-06-07, ending before the 2017-06-24 failure. It therefore cannot be used as direct evidence of event-related coherence loss.

Even as a pre-event coherence image, it shows the key limitation of this method in the study area: coherence is already low across broad parts of the mountainous landscape. A simple low-coherence threshold would identify many areas unrelated to the landslide. Coherence loss could still be useful if a true event-spanning interferogram were downloaded, but it would need to be interpreted together with topography, optical imagery, and backscatter change.

6 Sentinel-1 VV Backscatter Change

The notebook generated a true pre/post Sentinel-1 GRD comparison using VV backscatter. The pre-event composite contains acquisitions from 2017-05-02 to 2017-06-19, and the post-event composite contains acquisitions after 2017-06-24. Unlike the available local LiCSAR coherence raster, this product does compare imagery from before and after the failure.

VV backscatter change is sensitive to surface roughness, vegetation removal, exposed fresh debris, moisture, and changes in radar geometry. For this specific workflow, VV change is therefore a better SAR-based mapping product than the available coherence image, because it captures an actual pre/post-event surface change. It is still not perfect: radar shadows, layover, slope orientation, and moisture changes can create false positives. But for mapping the landslide footprint with the generated products, VV change is more directly relevant than the pre-event coherence file.

7 Discussion

The SAR results address the main SAR tasks from the lab. The LiCSAR interferograms were downloaded and converted to a MintPy stack, the subset was processed with SBAS, and the resulting velocity and displacement products show possible pre-failure deformation. The most useful evidence for slow motion is the localized cm/year-scale velocity pattern and the cumulative displacement visible by early 2017. These patterns are consistent with slope instability prior to the Xinmo landslide.

The main limitation is coherence. Temporal coherence and average spatial coherence are both patchy, which means that the InSAR time-series result is not reliable everywhere. The mountainous terrain likely causes radar shadow, layover, vegetation decorrelation, and seasonal effects. Because of this, the velocity field should be treated as an indicator of possible deformation rather than a clean, complete map of the unstable slope.

For mapping the landslide itself, the current coherence result is limited because the closest available local coherence raster ends on 2017-06-07. It does not span the 2017-06-24 event. Therefore, the saved coherence plot should not be claimed as a direct map of landslide coherence loss. A true event-spanning LiCSAR coherence product would be needed for that.

The VV backscatter-change workflow is better suited to mapping the event using the products generated here, because it compares Sentinel-1 GRD data before and after the landslide. The expected landslide signature would be a localized change in VV caused by vegetation removal, fresh debris exposure, and altered surface roughness. In the final interpretation, VV change should be used alongside optical imagery to refine the landslide extent.

8 Conclusion

The MintPy SBAS analysis provides useful evidence of pre-event deformation in the landslide area, with localized cm/year-scale line-of-sight motion and cumulative displacement before 2017-06-24. However, the reliability is limited by low and patchy coherence in the steep mountainous terrain. Coherence loss is not a strong mapping product from the current local files because the available coherence raster does not span the event and because low coherence occurs widely outside the landslide. Sentinel-1 VV backscatter change is the more useful SAR product for mapping the landslide footprint in this workflow, because it uses true pre- and post-event acquisitions.

9 Remaining Optical Task

The lab also asks for Sentinel-2 optical mapping using cloud-free pre- and post-event images and an index such as NDVI or BSI. No generated Sentinel-2 optical figures were present in the current output folders, so that section is not included as a completed result here. To finish the complete lab report, add cloud-free Sentinel-2 pre/post images, a mapped landslide extent, and an index-change result showing vegetation loss or bare-soil increase.